AI / ML Engineer · LLMs · RAG · Agentic Systems · MLOps

Jewin Cheng

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PROFILE

AI / ML Engineer with 2 years of experience building and deploying production AI systems at Malakoff Humanis. I make unstructured data usable — LLMs, RAG, agentic systems — with end-to-end ownership across NLP, computer vision and MLOps. Comfortable bridging technical delivery and business needs. Seeking a CDI as an AI / ML Engineer.

PROFESSIONAL EXPERIENCE

Malakoff Humanis — MH Tech, Data Science **Paris, France**
AI Engineer/Data Scientist 2024 – 2026

- Led ingestion for the **RAG service** behind an internal **HR assistant chatbot**: made the **scraping → chunking → indexing** chain (**BeautifulSoup**, 180 articles → 1K chunks) autonomous as a containerized job, added **content-hash incremental re-ingestion** cutting a **2-hour full re-index to minutes**, and fixed a **silent LLM truncation** bug dropping article content — **OpenSearch** hybrid search (**BM25** + **k-NN**), **S3**, **LiteLLM** proxy, tests and coverage gating a **Jenkins** image build.
- Built and deployed a **batch document-classification pipeline** (PDF and OCR parsing → labelling → training → prediction) reaching **85% accuracy** on a **1.1M-document corpus**, extracting **200K critical health-insurance contracts** and unblocking work that had been blocked for years for contract managers (8-month, 5-person project).
- Trained and integrated an **R-CNN computer-vision** model into a French tax-notice **NER** pipeline (100K notices/ year), lifting end-to-end performance by **+7 pp** over the QR-code-only baseline.
- Fine-tuned **BERT** for **hierarchical email triage** (3 levels, **200+ leaf classes**, 20K emails/month) reaching **70% accuracy** in production; surfaced edge cases and process-evasion patterns to the project owner via weekly business reviews — outcomes the team had no way of measuring before.

Formal Methods Laboratory (LMF) — Université Paris-Saclay **Orsay, France**
Research Intern Jun 2024 – Aug 2024

- Studied **Monero zero-knowledge proofs** and built a **transaction-path visualization tool** using a public contaminated-transaction dataset; presented findings to the academic supervisor and a cyber police officer.

EDUCATION

Université Paris-Saclay **Orsay, France**
MSc Computer Science for Data Science (Master ISD) 2024 – 2026

Université Paris-Saclay **Orsay, France**
BSc Computer Science and Mathematics — final year in parcours magistère (selective honors) 2021 – 2024

ACADEMIC PROJECTS

Hackathon Gen AI 2026 — Multi-agent code-generation VS Code extension
Paris-Saclay × Paris 8 × UQAC (team of 4) — 2026

- Built a **LangGraph** and **LangChain** multi-agent pipeline that turns a user need into specifications, validation, code and CI/CD configuration files, packaged as a **VS Code extension**.

RAG Ticket-Resolution Chatbot
MSc school project (team of 4) — 2025

- Designed a **Snowflake Cortex** (Agent, Search, REST API) and **LangChain / LangGraph** RAG system, deployed as a **Streamlit** app on Snowflake; achieved **70% LLM-judged faithfulness and groundedness** on 2K sample tickets.

RAG Nutrition Assistant
MSc course project — team lead (5 people) — 2025

- Led a **team of 5** on feature design (nutri-score computation, ingredient substitution, recipe rewriting) and coordinated with the project's other team leads; delivered a chatbot web application in 2 months.

SKILLS

Programming: Python, SQL, Java, OCaml, Rust, JavaScript / Node.js, C / C++
AI / ML: PyTorch, scikit-learn, Transformers (BERT), NER, computer vision, spaCy, NLTK, PyMuPDF, NumPy, Pandas
LLM / Agentic: LangChain, LangGraph, LiteLLM, RAG (chunking, embeddings, hybrid BM25 + k-NN retrieval), multi-agent systems, prompt engineering, fine-tuning, LLM evaluation (faithfulness / groundedness), Snowflake Cortex
Data / MLOps: MLflow, Docker / ECR, Kubernetes, Jenkins, CI/CD, Airflow, Spark, ETL / ingestion pipelines, AWS S3, AWS OpenSearch, Snowflake, Dataiku
Tools: Git / GitHub / GitLab, Linux, JupyterLab, Visual Studio Code, IntelliJ IDEA, Tableau
Certifications: Dataiku Core Designer; Dataiku ML Practitioner
Languages: French (native); Mandarin Chinese (native); English (TOEIC 925/990, Linguaskill 177/180, EUGLOH program)